FUNDAMENTALS OF DATA SCIENCE LAB

EXPERIMENT 5

Data Collection and Initial Exploration

Name: S. Amirtha Sharan

Register Number: 250701050

Exp No:5 Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. **Data Collection:**

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

```
[1] import pandas as pd
    db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

2. Initial Exploration:

```python
# Dataset structure and summary statistics
print(db.info())
print(db.describe())

# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

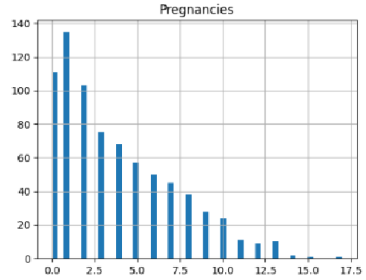
db.hist(bins=50, figsize=(20, 15))
plt.show()

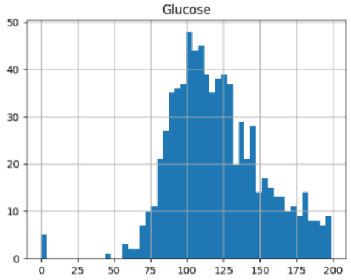
# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()
```

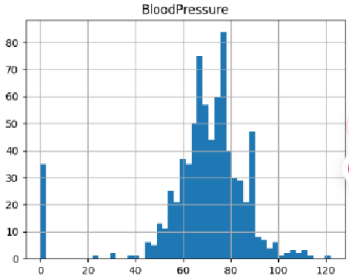
[3]

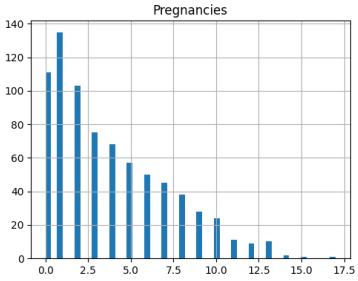
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Pregnancies               768 non-null    int64  
 1   Glucose                   768 non-null    int64  
 2   BloodPressure             768 non-null    int64  
 3   SkinThickness             768 non-null    int64  
 4   Insulin                   768 non-null    int64  
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64  
 8   Outcome                   768 non-null    int64  
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000   
mean      3.845052  120.894531      69.105469      20.536458   79.799479   
std       3.369578   31.972618      19.355807      15.952218  115.244002   
min       0.000000    0.000000       0.000000       0.000000    0.000000   
25%       1.000000   99.000000      62.000000       0.000000    0.000000   
50%       3.000000  117.000000      72.000000      23.000000   30.500000   
75%       6.000000  140.250000      80.000000      32.000000  127.250000   
max      17.000000  199.000000     122.000000      99.000000  846.000000   
```

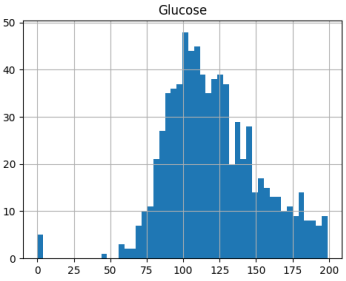
	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

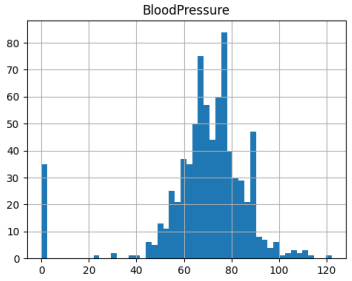


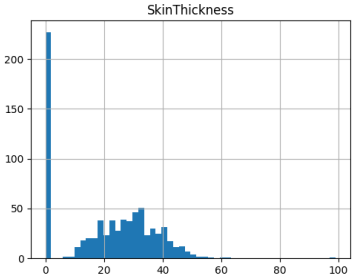


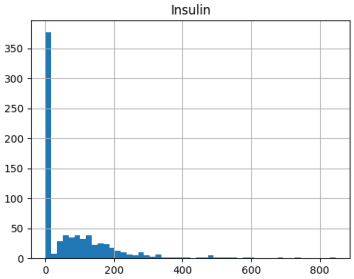


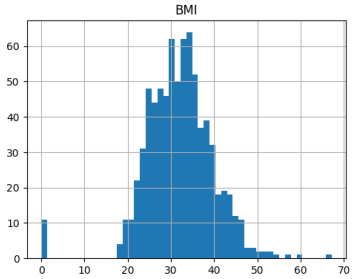


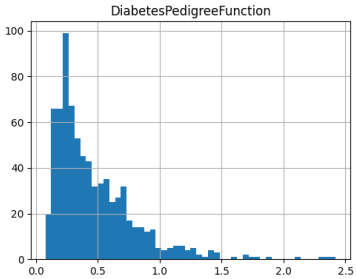


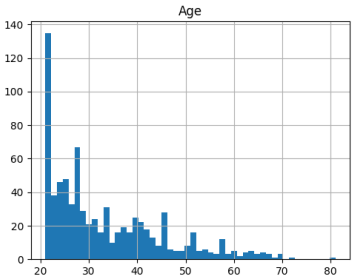


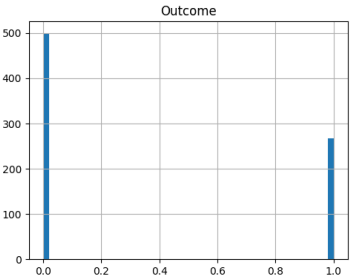


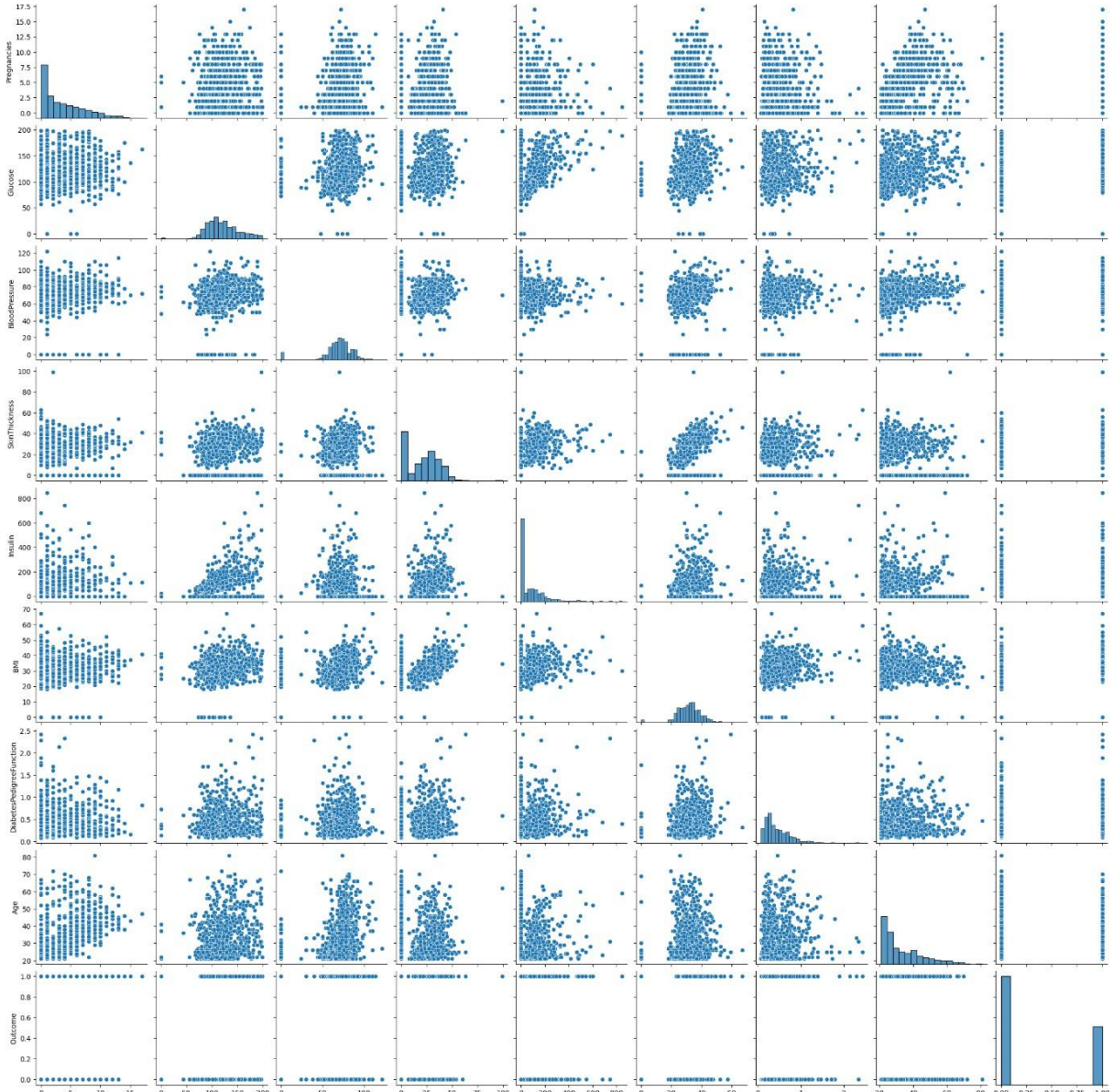











Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

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